CDA 106
Numerical Methods for Data Science

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Gram Schmidt Orthogonalization ($\mathbb{R}^n$)

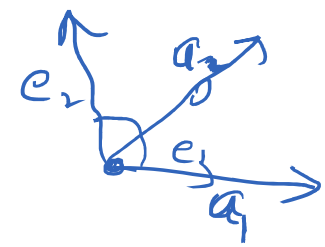
$$a_i = \begin{bmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{bmatrix}$$

vector $S = \{a_1, a_2, a_3, \ldots, a_n\}$

$\longrightarrow S_1 = \{u_1, u_2 \ldots u_n\}$ s.t. S_1 is orthogonal. i.e. $(u_i \perp u_j \ \forall i, j)$

$$e_1 = \frac{u_1}{\|u_1\|}$$ unit vector

Process $\longrightarrow u_1 = a_1$



$$\longrightarrow u_2 = a_2 - \underbrace{\langle a_2, e_1 \rangle}_{\text{project.}} \underbrace{e_1}_{\text{unit vector}}$$

$$\boxed{e_2 = \frac{u_2}{\|u_2\|}}$$

$$\langle a_2, e_1 \rangle = a_2^T e_1 = a_2^T \frac{u_1}{\|u_1\|} = \underbrace{\frac{a_2^T u_1}{\|u_1\|}}_{\text{projection of } a_2 \text{ on } u_1}$$

Claim $u_2 \perp u_1$ i.e. $\langle u_2, u_1 \rangle = 0$

equivalent $\langle u_2, e_1 \rangle = 0$

LHS

$$\langle u_2, e_1 \rangle = u_2^T e_1$$

$$= (a_2 - \underbrace{\langle a_2, e_1 \rangle}_{\text{scalar}} e_1)^T e_1$$

$$= (a_2^T - \langle a_2, e_1 \rangle e_1^T) e_1$$

$$= a_2^T e_1 - \langle a_2, e_1 \rangle e_1^T e_1$$

$$= \langle a_2, e_1 \rangle - \langle a_2, e_1 \rangle = 0$$

$$= \text{RHS}$$

$[(A-B)^T = A^T - B^T$

$e_1 \to$ unit vector

$\because e_1^T e_1 = 1$

because

$e_1^T e_1 = \|e_1\|^2$

$= 1$

By mistake I deleted slides of the class and saved file. Therefore I rewrote it. It may be different here & there.

Similarly,

$$\longrightarrow \quad u_3 = a_3 - \langle a_2, e_1 \rangle e_1 - \langle a_2, e_2 \rangle e_2$$

and the unit vector $\boxed{e_3 = \dfrac{u_3}{\|u_3\|}}$

<u>Claim</u> $u_3 \perp e_1$ & $u_3 \perp e_2$ (or $u_3 \perp u_1$ and $u_3 \perp u_2$)

<u>you can verify yourself.</u>

Continuing in similar manner

$$u_n = a_n - \langle a_n, e_1 \rangle e_1 - \langle a_n, e_2 \rangle e_2 - \cdots - \langle a_n, e_{n-1} \rangle e_{n-1}$$

and $\boxed{e_n = \dfrac{u_n}{\|u_n\|}}$

Therefore, $S = \{a_1, a_2, \ldots a_n\} \xrightarrow{\text{orthogonal}} \{u_1, u_2, \ldots u_n\}$

$$\{\langle u_i, u_j \rangle = 0 \quad i \neq j\}$$

$u_i \perp u_j \ \forall i, j$

In the process we get $\overbrace{\{e_1, e_2 \ldots e_n\}}^{\text{orthonormal}}$

s.t. $\langle e_i, e_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$.

Orthonormal Set

$S = \{a_1, \ldots, a_n\}$ is said to be orthonormal if

$$\langle a_i, a_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Therefore, in Gram Schmidt Orthogonalization process we get orthonormal Set.

Orthogonal matrix Q.

$$QQ^T = I = \begin{bmatrix} \ddots & 0 \\ 0 & \ddots \end{bmatrix}$$

entries are orthonormal set

<u>Inner product</u> $\langle a, b \rangle = a^T b$

$= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$

$a, b \in \mathbb{R}^n$

Example Using Gram Schmidt Orthogonalization process, find an orthonormal set $S_1 = \{e_1, e_2, e_3\}$ for the set

$$S = \left\{ a_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, a_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \; a_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Ans. We use Gram Schmidt Orthogonalization process.

choose $u_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ $\qquad \|a_1\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$

unit vector $e_1 = \dfrac{a_1}{\|a_1\|} = \dfrac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \boxed{e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}}$

Now, to find $u_2 = a_2 - \langle a_2, e_1 \rangle e_1$

$$\langle a_2, e_1 \rangle = a_2^T e_1 = \underbrace{\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}}_{a_2^T} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

Therefore, $u_2 = a_2 - \langle a_2, e_1 \rangle e_1$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix}$$

$$\underline{u_2 = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}}$$

$$\|u_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{3/2}$$

$$e_2 = \frac{u_2}{\|u_2\|} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix} \Rightarrow \boxed{e_2 = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ \sqrt{2/3} \end{bmatrix}}$$

$$u_3 = a_3 - \underline{\langle a_3, e_1 \rangle} e_1 - \underline{\langle a_3, e_2 \rangle} e_2$$

$$\langle a_3, e_1 \rangle = a_3^T e_1 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

and.

$$\underline{\langle a_3, e_2 \rangle} = a_3^T e_2 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ \sqrt{2/3} \end{bmatrix} = -\frac{1}{\sqrt{6}} + \sqrt{\frac{2}{3}} = \underline{\underline{\frac{1}{\sqrt{6}}}}$$

Therefore,

$$u_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} - \frac{1}{\sqrt{6}} \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ \sqrt{2/3} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/6 \\ -1/6 \\ 1/3 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix} \qquad \|u_3\| = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{4}{9}} = \frac{2\sqrt{3}}{3} = 2/\sqrt{3}$$

$$e_3 = \frac{u_3}{\|u_3\|} = \frac{\sqrt{3}}{2} \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix} \Rightarrow \boxed{e_3 = \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}}$$

$$S = \{a_1, a_2, a_3\} \xrightarrow[\text{set}]{\text{Orthonormal}} \{e_1, e_2, e_3\}$$

$$\langle e_i, e_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

$e_1 \perp e_2$, $e_2 \perp e_3$ & $e_1 \perp e_3$ (<u>you can verify</u>).

$\underline{\langle e_1, e_2 \rangle = 0}$ $\underline{\langle e_2, e_3 \rangle = 0}$ $\underline{\langle e_1, e_3 \rangle = 0}$

$$\langle e_2, e_3 \rangle = e_2^T e_3 = \begin{bmatrix} \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \sqrt{\frac{2}{3}} \end{bmatrix} \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} = -\frac{1}{3\sqrt{2}} - \frac{1}{3\sqrt{2}} + \frac{\sqrt{2}}{3}$$

$$= -\frac{2}{3\sqrt{2}} + \frac{\sqrt{2}}{3} = 0$$

QR Decomposition

$$A_{3\times 3} = \left[\, a_1 \,\middle|\, a_2 \,\middle|\, a_3 \,\right]$$

$$A = QR = \underbrace{\left[\, e_1 \,\middle|\, e_2 \,\middle|\, e_3 \,\right]}_{Q} \underbrace{\begin{bmatrix} a_1^T e_1 & a_2^T e_1 & a_3^T e_1 \\ 0 & a_2^T e_2 & a_3^T e_2 \\ 0 & 0 & a_3^T e_3 \end{bmatrix}}_{R}$$

without proof

$\{e_1, e_2, e_3\}$ — orthonormal vectors of $\{a_1, a_2, a_3\}$

Refer to Kreyszig / Strang.

Example Write $A = \begin{array}{c} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \\ \begin{array}{ccc} a_1 & a_2 & a_3 \end{array} \end{array} = QR.$

$S = \{a_1, a_2, a_3\} \xrightarrow{\text{orthonormal}} \{\underline{e_1}, \underline{e_2}, \underline{e_3}\}$ [See from slide 8 — 10]

From previous example, $e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ \sqrt{2/3} \end{bmatrix}$ & $e_3 = \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$.

Now, $a_1^T e_1 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \sqrt{2}$, $a_2^T e_2 = \langle a_2, e_2 \rangle = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ \sqrt{2/3} \end{bmatrix}$

$\boxed{a_2^T e_2 = \sqrt{3/2}}$

$a_2^T e_1 = \langle a_2, e_1 \rangle = \frac{1}{\sqrt{2}}$

$a_3^T e_2 = \frac{1}{\sqrt{6}}$

$a_3^T e_1 = \langle a_3, e_1 \rangle = \frac{1}{\sqrt{2}}$

$a_3^T e_3 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} = \frac{2}{\sqrt{3}}$

$$
\underbrace{\begin{array}{c} \begin{array}{ccc} a_1 & a_2 & a_3 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \end{array}}_{A}
=
\underbrace{\begin{array}{c} \begin{array}{ccc} e_1 & e_2 & e_3 \end{array} \\ \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \sqrt{2/3} & 1/\sqrt{3} \end{bmatrix} \end{array}}_{\underline{Q}}
\underbrace{\begin{array}{c} \begin{array}{ccc} a_1^T e_1 & a_2^T e_1 & a_3^T e_1 \end{array} \\ \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix} \end{array}}_{\underline{R}}
$$

Now, we can provide QR decomposition of any $n \times k$ matrix

$$\underline{\underline{A_{n\times k}}} = \left[\, a_1 \,\middle|\, a_2 \,\middle|\, \cdots \,\middle|\, a_k \,\right]_{n\times k} = \underbrace{\left[\, e_1 \,\middle|\, \cdots \,\middle|\, e_k \,\right]_{n\times k}}_{Q} \underbrace{\begin{bmatrix} a_1^T e_1 & a_2^T e_1 & \cdots & a_k^T e_1 \\ 0 & a_2^T e_2 & \cdots & a_k^T e_2 \\ \vdots & & & \\ 0 & \cdots 0 & \cdots & a_k^T e_k \end{bmatrix}_{k\times k}}_{R}$$

$$a_i = \begin{bmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{bmatrix} \quad \forall i \qquad S = \{a_1 \cdots a_k\} \xrightarrow{\text{orthonormal}} \{e_1 \cdots e_k\}$$

$$A_{n\times k} = \underline{\underline{B_{n\times k}}} \; C_{k\times k}$$

Note $R_{ij} = a_j^T e_i$

Example

$$A = \begin{array}{c} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}_{3\times 2} \\ \begin{array}{cc} a_1 & a_2 \end{array} \end{array}$$

Find QR decomposition!

$S = \{a_1, a_2\}$ $\xrightarrow[\text{orthogonalization}]{\text{Gram Schmidt}}$ $S_1 = \{e_1, e_2\}$

$$e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

$$e_2 = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \sqrt{2/3} \\ 1/\sqrt{6} \end{bmatrix}$$

Choose $u_1 = a_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ $\quad \|a_1\| = \sqrt{2}$

$$u_2 = a_2 - \langle a_2, e_1 \rangle e_1$$

$$\langle a_2, e_1 \rangle = a_2^T e_1 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{\sqrt{2}}$$

$$u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1 \\ 1/2 \end{bmatrix}$$

$$\|u_2\| = \sqrt{\frac{1}{4} + 1 + \frac{1}{4}} = \sqrt{3/2}$$

$$A_{3\times 2} = \left[a_1 \middle| a_2\right] = \underbrace{\left[e_1 \middle| e_2\right]_{3\times 2}}_{Q} \underbrace{\begin{bmatrix} a_1^T e_1 & a_2^T e_1 \\ 0 & a_2^T e_2 \end{bmatrix}_{2\times 2}}_{R}$$

$$a_1^T e_1 = \langle a_1, e_1 \rangle = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \sqrt{2}$$

$$a_2^T e_2 = \langle a_2, e_2 \rangle = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \sqrt{2/3} \\ 1/\sqrt{2} \end{bmatrix} = \sqrt{\frac{3}{2}}$$

<u>Therefore</u>

$$\underbrace{\begin{array}{c} \begin{array}{cc} a_1 & a_2 \end{array} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}_{3\times 2} \end{array}}_{A} = \underbrace{\begin{array}{c} \begin{array}{cc} e_1 & e_2 \end{array} \\ \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ 0 & \sqrt{2/3} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix}_{3\times 2} \end{array}}_{Q} \underbrace{\begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{\frac{3}{2}} \end{bmatrix}_{2\times 2}}_{R}.$$

How shall we use QR decomposition

$$Ax = b \qquad \text{sol}^n \quad \underline{x = ?}$$

$$\Rightarrow QRx = b$$

$$\Rightarrow Rx = Q^{-1}b = Q^T b = \tilde{b}$$

$$\Rightarrow \underline{Rx = \tilde{b}}$$

Backward substitution

$$\rightarrow QQ^T = I$$

$$\underline{Q^{-1} = Q^T}$$

$\underline{Q \rightarrow \text{square}}$

$\rightsquigarrow$ $\underline{\text{Tilde}}$

$$x = ?$$

$$\boxed{Ax = b} \xrightarrow[\text{decom.}]{QR} \boxed{Rx = Q^T b}$$

To solve $Ax = b$ $\longrightarrow$ When the system has sol^n.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \qquad X = \begin{bmatrix} x \\ y \end{bmatrix} \qquad b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$A = QR$$

[We already did QR decomposition of A Refer sl. 17]

To solve $Ax = b \Rightarrow QRx = b$

$$\Rightarrow Rx = Q^T b$$

$$\begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -\frac{1}{\sqrt{6}} & \sqrt{\frac{2}{3}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ \sqrt{2/3} \end{bmatrix}$$

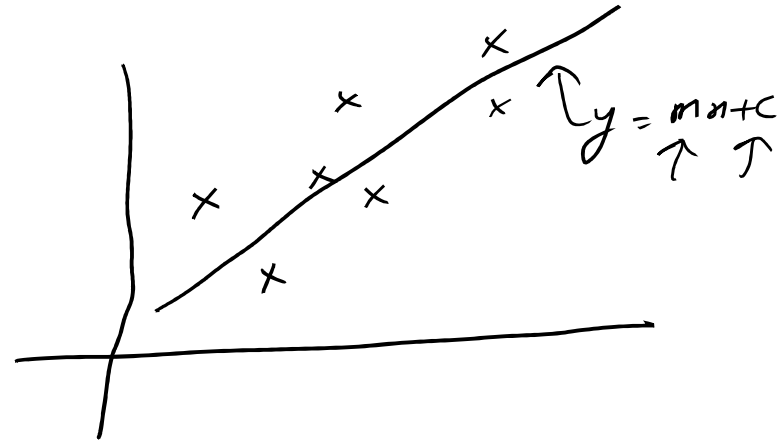
$$\sqrt{2}\,x + \frac{1}{\sqrt{2}}\,y = \sqrt{2} \Rightarrow 2x + y = 2$$

Backward sub.

$$y = 2/3 \qquad x = \frac{2}{3}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \left.\begin{array}{l} x = 1 \\ y = 1 \\ x+y = 1 \end{array}\right\}$$

We need to carefully understand our values of x, y which we got for $\underline{\underline{Ax = b}}$

Curve fitting / data fitting

<u>Least square</u>